CW-ERM: Improving Autonomous Driving Planning with Closed-loop Weighted Empirical Risk Minimization [1]

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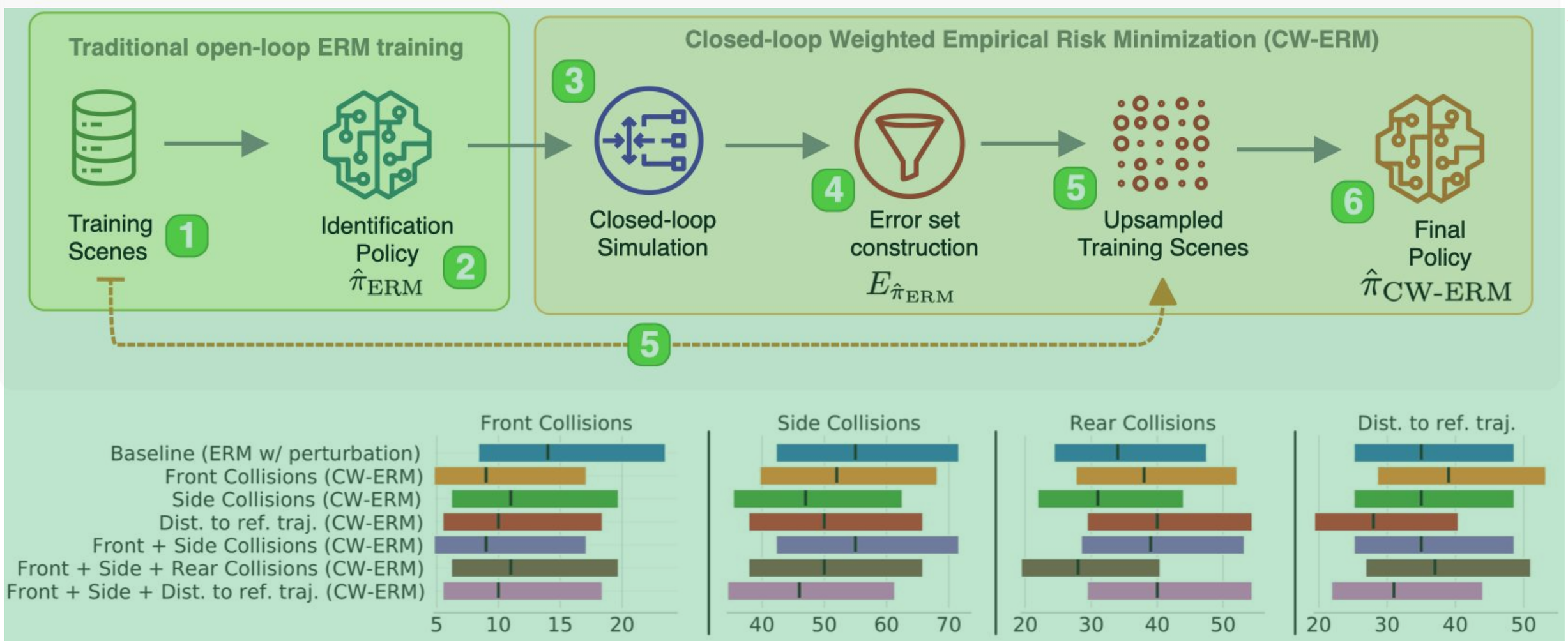
Main Ideas and Motivation [4]

- Behavior Cloning (BC) algorithms for training self-driving vehicles via pure Empirical Risk Minimization (ERM) biases policy towards matching open-loop behavior [5]
- We debias the policy network by identifying important samples via a closed-loop evaluator [6]
- Training in three phases [7]
  - Open-loop identification phase
  - Error set construction and evaluation phase
  - Upsampling training phase

Our Contributions [8]

- Motivate and propose a technique that leverages closed-loop evaluation metrics acquired from policy rollouts in a simulator to debias the policy network and reduce the distributional differences between training (open-loop) and inference time (closed-loop); [9]
- Evaluate CW-ERM on a challenging urban driving dataset in a closed-loop fashion to show that our method, although simple to implement, yields significant improvements in closed-loop performance without requiring complex and computationally expensive closed-loop training methods; [10]
- Show an important connection of our method to a family of methods that addresses covariate shift through density ratio estimation. [11]



Methodology [13]

Stage 1 (Identification Policy) [14]

- Train policy network via BC in open-loop using ERM, get $\hat{\pi}_{ERM}$ [15]

Stage 2 (Closed-Loop Simulation) [17]

- Perform rollouts using $\hat{\pi}_{ERM}$ in closed-loop simulator [18]
- Collect closed-loop metrics and identify the error set [19]

$$E_{\hat{\pi}_{ERM}} = \{(s_i \,.\, a_i) \text{ s.t. } C(s_i, a_i) > 0\}, \quad [20]$$

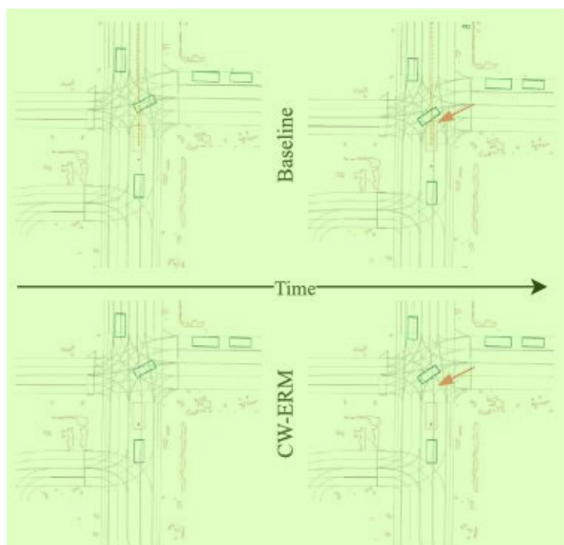
where $C(s_i, a_i)$ is a cost (e.g. collisions) incurred during the closed-loop rollouts. [21]

Stage 3 (Final Policy) [27]

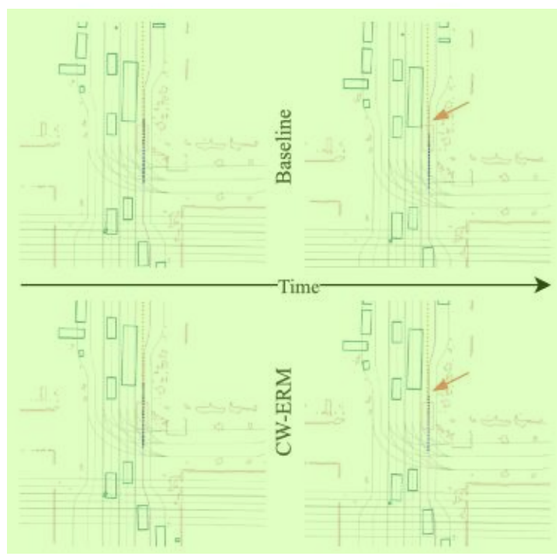
- Train new policy where scenes belonging to $E_{\hat{\pi}_{ERM}}$ are unweighted by a factor $w(\cdot)$. [28]

$$\arg\min_{\pi \in \Pi} \mathbb{E}_{s \sim d_{\pi^*}, a \sim \pi^*(s)}[w(E_{\hat{\pi}_{ERM}}, s)\ell(s, a, \pi)] \quad [29]$$

Collision Avoidance Examples [16]


 [22]

Front collisions [23]


 [24]

Side collisions [25]

References [26]

1. Liu et al. (2021). Just Train Twice: Improving Group Robustness without Training Group Information. *International Conference on Machine Learning*.
2. Vitelli et al. (2022). Safetynet: Safe planning for real-world self-driving vehicles using machine-learned policies. *International Conference on Robotics and Automation (ICRA)*.
3. Scheel et al. (2022). Urban driver: Learning to drive from real-world demonstrations using policy gradients. *Conference on Robot Learning*. [30]

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